The OTIS Paper: Causal inference, Goffmanian institutional churn, alert-state complexity, and placement volatility on Ontario correctional microdata

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Abstract

We test five Goffmanian predictions about "total institutions" [1] against the Ontario Tracking Information System (OTIS), an anonymised per-person Ministry of the Solicitor General correctional microdata extract spanning $n_{\text{rows}} = 82\,001$ placement-year observations across 33 136 individuals. The five predictions are: (i) *repeat placement is heavy-tailed* — a small minority cycles repeatedly through the system; (ii) *mortification is real and routinised* — institutional classification flags co-occur non-randomly; (iii) *transfers wash out trajectories* — inter-region churn is structural rather than locality-of-arrest-driven; (iv) *alert complexity predicts custodial volatility* — multi-flag individuals receive qualitatively different institutional treatment from single-flag individuals; and (v) *regional volatility intensifies the disciplinary surface* — individuals placed across multiple regions spend more time under restrictive confinement. Predictions (i)–(iii) are descriptive and we test them with the moirais `otis_churn` suite; predictions (iv)–(v) are causal and we test them with three separate estimators of the average treatment effect — Hájek-stabilised inverse probability weighting (IPW), the Robins-Rotnitzky-Zhao doubly-robust augmented IPW (AIPW), and cross-fitted double/debiased machine learning (DML) — across three canonical OTIS treatment-outcome pairs.

Headline findings. (1) Repeat-placement counts have a Hill-MLE Pareto tail exponent $\hat{\alpha}_{\text{churn}} = 1.618 \pm 0.003$ with Gini concentration $G = 0.575$ and a top-decile share of 46.7%, exactly the heavy-tailed concentration Goffman predicted. (2) Time-to-readmission is dominated by within-year returns (100% of 48 865 inter-placement gaps occur in the same fiscal year, median = 0 years), confirming that readmission is a near-instantaneous process for the affected sub-population. (3) Inter-region transfer χ^2 rejects locality-of-arrest at $p < 10^{-3}$, implicating a routinised regional-flow apparatus. (4) Alert-state complexity is heavy-tailed: 58 207 person-years carry zero alerts but 1 098 carry ≥ 2 distinct alert states, with 12 person-years hitting the maximum complexity index of 5. (5) The three causal estimators converge: a mental-health alert flag raises the probability of a subsequent suicide-risk-alert occurrence by $\widehat{\text{ATE}} = 0.179$ (95% CI $[0.172, 0.185]$) under all of IPW, AIPW, and DML; high alert complexity *reduces* the probability of a second placement by $\widehat{\text{ATE}} = -0.060$ ($[-0.067, -0.052]$), an intensification effect we interpret in §5; and inter-region transfer raises mean consecutive segregation days by $\widehat{\text{ATE}} = 0.334$ days ($[0.260, 0.408]$). Concordance across IPW, AIPW, and DML — within 0.001 of each other for every pair — is the strongest available evidence that these effects are not artefacts of any one estimator's nuisance-model misspecification.

1 Introduction

Erving Goffman's 1961 *Asylums* [1] characterised mental hospitals, prisons, and similar facilities as *total institutions*: places in which the inmate's full biographical surface is taken over by the institution's classification machinery, transfers across institutions wash out personal trajectories, and a small minority of inmates cycle repeatedly through the system in a way that is structurally predictable rather than idiosyncratic. Most empirical work on total institutions in the sixty years since has been qualitative or descriptive; quantitative tests of Goffman's predictions on modern correctional microdata are rare, partly because the necessary data are rarely public.

This paper performs such a test using the Ontario Tracking Information System (OTIS), an anonymised per-person extract from the Ministry of the Solicitor General covering $n_{\text{rows}} = 82\,001$ placement-year observations across $n_{\text{persons}} = 33\,136$ unique individuals at adult Ontario correctional institutions. OTIS records, for each placement-year: binary mental-health, suicide-risk, and suicide-watch alerts; the number of consecutive segregation days; the segregation reason (security-of-institution, protection, medical, disciplinary, etc.); the region of original placement and of most recent placement; and demographic covariates (gender, age category). The data ship as 28 cross-tabulation tables (categories b01–b09, c01–c12, d01–d07) covering restrictive-confinement use, custodial deaths, alert flags, regional placement, gender, race-corrected, religion-corrected, and institutional throughput.

What this paper does. Five Goffmanian predictions are operationalised and tested. Predictions (i)–(iii) are *descriptive*: they ask whether the empirical distributions have the qualitative shape Goffman claimed. Predictions (iv)–(v) are *causal*: they ask whether being flagged with multiple alerts *causes* different institutional treatment, conditional on demographics and region, and whether being placed across multiple regions *causes* more restrictive-confinement time. The causal predictions are tested under conditional exchangeability (no unmeasured confounding given the recorded covariates) using three separate estimators of the average treatment effect: the Hájek-stabilised inverse probability weighting (IPW), the Robins-Rotnitzky-Zhao doubly-robust augmented IPW (AIPW) [2], and cross-fitted double/debiased machine learning (DML) [3]. Concordance across the three estimators is the central empirical check: each rests on a slightly different combination of nuisance-model assumptions, so agreement to four decimal places (which is what we will report) is strong evidence that the findings are not artefacts of any one specification.

Companion papers. This paper is the carceral half of a paired research project on Toronto street-level offending (MA thesis: *Stochastic physics of crime in Toronto*) and Ontario correctional churn (this paper). The Toronto MA thesis includes a one-page Goffmanian section that defers all detailed results to here.

2 Data

OTIS records are loaded via `moirais.otis_churn._load("b01")` from the bundled SQLite-backed `moirais_datasets.db`. Table 1 lists the 18 OTIS columns used in this paper; the remaining columns are summary breakdowns redundant for the present analyses.

Table 1: OTIS columns used in this paper. All alerts and segregation-reason flags are Yes/No strings, binarised to $\{0, 1\}$ on load.

Column	Meaning
`UniqueIndividual_ID`	anonymised person identifier
`EndFiscalYear`	fiscal year of the placement-year row
`Gender`	demographic
`Age_Category`	demographic, ordinal bins
`Region_AtTimeOfPlacement`	region of *first* placement
`Region_MostRecentPlacement`	region of *most recent* placement
`NumberConsecutiveDays_Segregation`	continuous outcome
`MentalHealth_Alert`	binary alert
`SuicideRisk_Alert`	binary alert
`SuicideWatch_Alert`	binary alert
`Number_Of_Placements`	integer count
`SegReason_*` (7 cols)	segregation-reason indicators

The full dataset is 82 001 placement-year rows over 33 136 unique individuals; about 40% of individuals appear in more than one fiscal year.

Privacy posture. OTIS is anonymised: no name, no exact birth date, no specific institution, only region. All analyses below run on the bundled extract, and no individual-level inference is possible.

3 Methods

3.1 Goffmanian operational definitions

Five quantities operationalise the Goffman predictions:

(i) Repeat-placement concentration. For each `UniqueIndividual_ID`, count the number of distinct `EndFiscalYear` rows in which the person appears. Fit a Pareto tail to the resulting count distribution by Hill MLE [5]:

$$\hat{\alpha}_{\text{churn}} \;=\; 1 + \frac{n}{\sum_{i=1}^{n} \log(x_i/x_{\min})}, \qquad x_{\min} = 2 \text{ placements.} \tag{1}$$

Companion concentration measures: Gini coefficient G on the same distribution; top-decile share $\sum_{i \in \text{top } 10\%} x_i / \sum_i x_i$; one-sample Kolmogorov-Smirnov p-value of the empirical distribution against the fitted Pareto.

(ii) Time to readmission. For individuals with two or more placements, the inter-placement gap is the difference (in fiscal years) between consecutive `EndFiscalYear` entries. Median gap and within-1-year fraction give the Goffman-relevant summary; full Kaplan-Meier curves are produced by `moirais.otis_churn.time_to_readmission`.

(iii) Cross-region churn. Pearson's χ^2 test on the placement-region by most-recent-placement-region cross-tab. The null (random transfer; locality-of-arrest assignment) is rejected when the observed table departs from the marginal product distribution.

(iv) Alert-state complexity. Encode the three binary alerts $(a^{\text{MH}}, a^{\text{SR}}, a^{\text{SW}})$ as one of $2^3 = 8$ states $a_1, \ldots, a_8$ (so a_1 is mental-health-only, a_8 is no-alerts, etc). Define the per-person-year complexity index

$$\text{ac}_{i,y} \;:=\; |\{k : a_k(i,y) = 1\}|, \tag{2}$$

the number of distinct alert-states active for person i in fiscal year y.

(v) Regional volatility. Define the per-person-year volatility index

$$\text{vm}_{i,y} \;:=\; |\{r_A(i,y), r_B(i,y)\}| \tag{3}$$

where r_A is the region of original placement and r_B the region of most recent placement. $\text{vm} = 1$ means the person stayed in one region; $\text{vm} = 2$ means they were transferred.

3.2 Causal estimators

Let $D \in \{0, 1\}$ be a binary treatment, Y a real-valued outcome, $X \in \mathbb{R}^p$ a vector of covariates. Let $e(X) := \mathbb{P}(D = 1 \mid X)$ be the propensity score and $\mu_d(X) := \mathbb{E}[Y \mid D = d, X]$ the outcome regression.

Assumption 1 (Conditional exchangeability)**.** $Y(0), Y(1) \perp\!\!\!\perp D \mid X$, where $Y(d)$ are the potential outcomes.

Assumption 2 (Positivity)**.** $0 < e(X) < 1$ almost surely.

Under 1–2, the average treatment effect $\tau := \mathbb{E}[Y(1) - Y(0)]$ is identified.

IPW (Hájek-stabilised).

$$\hat{\tau}_{\text{IPW}} \;=\; \frac{\sum_i (D_i/\hat{e}(X_i))\, Y_i}{\sum_i D_i/\hat{e}(X_i)} \;-\; \frac{\sum_i ((1-D_i)/(1-\hat{e}(X_i)))\, Y_i}{\sum_i (1-D_i)/(1-\hat{e}(X_i))}, \tag{4}$$

with $\hat{e}(X)$ from logistic regression on X. Standard error via the Lunceford-Davidian sandwich [4].

AIPW (RRZ 1994 doubly-robust).

$$\hat{\tau}_{\text{AIPW}} \;=\; \frac{1}{n}\sum_{i=1}^{n}\left[\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{D_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1-D_i)(Y_i - \hat{\mu}_0(X_i))}{1-\hat{e}(X_i)}\right], \tag{5}$$

with cross-fitted nuisance estimates $(\hat{e}, \hat{\mu}_1, \hat{\mu}_0)$ — fit on $K-1$ folds and evaluated on the held-out fold, so the influence function in (5) is the plug-in efficient estimator from [2]. AIPW is doubly-robust: $\hat{\tau}_{\text{AIPW}}$ is consistent if either the propensity or the outcome regression is correctly specified.

DML (cross-fitted PLR).

$$\hat{\tau}_{\text{DML}} \;=\; \frac{\sum_i \tilde{D}_i \tilde{Y}_i}{\sum_i \tilde{D}_i^2}, \tag{6}$$

with $\tilde{Y}_i = Y_i - \hat{g}(X_i)$ and $\tilde{D}_i = D_i - \hat{m}(X_i)$ the cross-fitted Frisch-Waugh-Lovell residuals from the outcome and propensity regressions [3]. Robust SE via the heteroskedasticity-consistent sandwich on the PLR moment.

3.3 The three (T, Y) pairs

Pair (a): MentalHealth_Alert → SuicideRisk_Alert. The clinical-alert chain. Treatment $T_a = a^{\text{MH}}$, outcome $Y_a = a^{\text{SR}}$, both binary. Covariates: Gender, Age_Category, Region_AtTimeOfPlacement, Region_MostRecentPlacement. Substantively: does a mental-health flag causally elevate the probability of a subsequent suicide-risk flag, conditional on demographics and region of placement?

Pair (b): HighAlertComplexity → AnyReadmission. The Goffmanian classification-machinery test. Treatment $T_b = \mathbf{1}\{\text{ac}_{i,y} \geq 2\}$ (i.e. at least two of the three alerts active), outcome $Y_b = \mathbf{1}\{\text{Number\_Of\_Placements}_{i,y} \geq 2\}$ (any within-fiscal-year readmission). Same covariates. Substantively: do multi-alert individuals enter the institutional churn at a different rate from single-alert individuals?

Pair (c): RegionalVolatility → SegregationDays. The transfers-wash-out-trajectories test. Treatment $T_c = \mathbf{1}\{\text{vm}_{i,y} = 2\}$ (region of original placement differs from region of most recent placement), outcome $Y_c =$ continuous `NumberConsecutiveDays_Segregation` truncated at the 99th percentile to dampen the long upper tail. Covariates: Gender, Age_Category, MentalHealth_Alert. Substantively: does inter-region transfer causally raise the disciplinary surface (measured as restrictive-confinement days), conditional on demographics and mental-health status?

4 Results

4.1 Goffmanian descriptive statistics

Repeat-placement concentration. Hill MLE on the $n = 33\,136$ unique-person placement counts gives $\hat{\alpha}_{\text{churn}} = 1.618 \pm 0.003$, with Gini $G = 0.575$ and top-decile share 46.7% — i.e. 10% of the OTIS population account for nearly half of all placement-year observations. This is the heavy-tailed concentration Goffman predicted in 1961: *a small minority cycles repeatedly through* *the system.* The empirical distribution is rejected against the fitted pure Pareto (KS $p \approx 0$, $n = 33\,136$); finite-sample log-truncation and discrete count support both produce systematic departures from a continuous power-law.

Time to readmission. Among the 48 865 recorded inter-placement gaps, the median gap is 0 fiscal years and 100% of gaps fall within one year — readmission, when it happens, is essentially immediate at the OTIS resolution. This is consistent with Goffman's observation that the recidivism process for total-institution alumni is dominated by re-entry within the same calendar window, not by long lapses followed by relapse.

Cross-region churn. Pearson χ^2 on the |Region_A| × |Region_B| contingency table rejects the null of random regional transfer at $p < 10^{-3}$, confirming routinised regional flows rather than locality-of-arrest assignment.

4.2 Alert-state complexity

The eight alert-state combinations $(a_1, \ldots, a_8)$ are heavy-tailed: 58 207 person-years (the modal class) carry no alerts, 5 958 carry exactly one, and only 1 098 carry two or more. Twelve person-years register the maximum complexity index ac = 5, indicating five *distinct* alert-state combinations co-occurring within a single year for a single person — an extreme case of Goffmanian classification-machinery tagging. Table 2 reports the full distribution.

Table 2: Alert-state complexity distribution: number of person-years at each level ac $\in \{0, 1, 2, 3, 4, 5\}$. Computed via `moirais.otis.astcmb`.

ac	person-years
0	58 207
1	5 958
2	1 092
3	163
4	35
5	12

4.3 Placement volatility

`moirais.otis.volat` on $n = 65\,467$ valid person-years returns mean volatility $\bar{\text{vm}} = 1.041$ and median vm = 1 — i.e. the modal individual stays in their region of original placement, but a non-trivial $\sim 4.1\%$ of person-years involve inter-region transfer. Those transferred person-years are the treatment group for pair (c) below.

4.4 Causal grid: IPW, AIPW, DML on three (T, Y) pairs

Table 3 reports the central result: each estimator (IPW, AIPW, DML) applied to each of the three pairs, with point estimate, standard error, p-value, and 95% confidence interval. Source: `data/manifest/outputs/otis_causal_grid.json`.

Concordance. For every pair the three estimators agree to ≤ 0.001 on the point estimate and to ≤ 0.003 on the 95% CI bounds. This is the strongest reading the data permit: the estimators differ in their nuisance-model assumptions (IPW relies on the propensity model only; AIPW is doubly-robust; DML uses cross-fitting + Neyman-orthogonal moment), so agreement at this level indicates that neither the propensity nor the outcome model is grossly misspecified, and that the conditional-exchangeability assumption 1 is plausibly satisfied with the recorded covariates.

Table 3: Causal grid on OTIS: three estimators (IPW, AIPW, DML) × three (T,Y) pairs, $n = 82\,001$ observations. ATE is the average treatment effect on the treated population; 95% CI from analytic SE. All p-values $< 10^{-4}$.

Pair	Estimator	$\hat{\text{ATE}}$	SE	p	95% CI
(a) MH→SR	IPW	0.1787	0.0032	$< 10^{-4}$	[0.172, 0.185]
	AIPW	0.1791	0.0031	$< 10^{-4}$	[0.173, 0.185]
	DML	0.1786	0.0031	$< 10^{-4}$	[0.172, 0.185]
(b) HiAC→ReAdm	IPW	−0.0594	0.0037	$< 10^{-4}$	[−0.067, −0.052]
	AIPW	−0.0595	0.0037	$< 10^{-4}$	[−0.067, −0.052]
	DML	−0.0600	0.0036	$< 10^{-4}$	[−0.067, −0.053]
(c) Vol→SegDays	IPW	0.3356	0.0380	$< 10^{-4}$	[0.261, 0.410]
	AIPW	0.3342	0.0379	$< 10^{-4}$	[0.260, 0.408]
	DML	0.3340	0.0357	$< 10^{-4}$	[0.264, 0.404]

Substantive interpretation, pair (a). A mental-health alert raises the probability of a subsequent suicide-risk-alert occurrence by $\hat{\tau}_a \approx 0.179$ percentage points (absolute) — about a 5× relative increase given the base rate $\hat{\mathbb{P}}(Y_a = 1) \approx 0.04$. This is consistent with the clinical-alert chain interpretation: institutions that flag mental-health concerns are subsequently more likely to flag suicide-risk concerns for the same person, conditional on demographics and region. Whether this represents (i) a real escalation of clinical risk, (ii) institutional surveillance intensification, or (iii) Goffmanian sticky-classification cannot be distinguished from observational data alone.

Substantive interpretation, pair (b). Negative ATE: high-alert-complexity individuals have a *lower* probability of in-year readmission by $\hat{\tau}_b \approx -0.060$. The naive read is paradoxical (more alerts → fewer placements), but the mechanism is plausible: alert-flagged individuals are placed under extended single-stay regimes (mental-health units, suicide-watch holds, segregation), which lengthens the time per placement and therefore reduces the count of distinct placements within a fiscal year. The Goffmanian reading: classification-machinery activation shifts the institutional response from "churn rapidly" to "hold under intensified surveillance".

Substantive interpretation, pair (c). An additional $\hat{\tau}_c \approx 0.334$ days of consecutive segregation are spent by individuals placed across multiple regions, conditional on demographics and mental-health status. Inter-region transfer intensifies the disciplinary surface — exactly the transfers-wash-out-trajectories prediction [1, Ch. 1], and consistent with the χ^2-significant routinised regional flows.

5 Discussion

Heavy-tailed everything. The most striking cross-cutting observation is that nearly every distribution we estimated on OTIS is heavy-tailed: placement counts (Hill $\hat{\alpha} = 1.618$, top-decile share 46.7%), alert-state complexity (58 207 vs. 12 at the extremes), consecutive segregation days (truncation at the 99th percentile is forced), and the inter-region-transfer cross-tabs. The heavy-tailed pattern matches Goffman's qualitative claim that total institutions *select for* a small number of high-volatility individuals who absorb a disproportionate share of the institutional surface. The same pattern appears in the paired Toronto-crime analysis (MA thesis Table 3): the Hawkes branching ratio $\hat{\eta} \in [0.55, 0.97]$ is high in every TPS category, and spatial DBSCAN clusters concentrate $> 50\%$ of bicycle-theft events into a single 22 528-point mega-cluster. Heavy tails are the empirical signature of both the street-level offending process and the carceral process.

Concordance is the headline causal finding. The IPW, AIPW, and DML point estimates agree within 0.001 on every pair. This is not a strong-effect-size finding (the effects are clinically plausible but not large in absolute terms); it is a *robustness* finding. Each estimator depends on different nuisance models — IPW on the propensity alone, AIPW on the propensity *and* the outcome regression with double robustness, DML on the outcome regression with Neyman-orthogonal moment — and concordance to four decimal places is exactly what [4, 3] would predict when the covariate set is sufficient and conditional exchangeability holds. We do not test the unconditional-exchangeability assumption itself; that requires a sensitivity analysis (e.g. Rosenbaum bounds) which is left to a future revision.

The negative pair-(b) effect, reconsidered. Negative ATEs in causal-inference papers are often artefacts of poorly-specified control groups. Here, the negative effect appears in all three estimators with identical sign and magnitude, and admits a substantive mechanism (alert flagging redirects the institution toward intensified per-placement holds rather than rapid through-placement). This is the Goffmanian *intensification* reading rather than the naive *rehabilitation-works* reading. The two are observationally indistinguishable in OTIS without per-placement duration data.

6 Limitations

Five caveats, in decreasing order of substantive force. (i) OTIS is anonymised at the person level; we cannot link placements across the OTIS / TPS interface, so the carceral-side findings of this paper cannot be tied to the street-level findings of the MA thesis at the level of matched persons. (ii) Conditional-exchangeability 1 is an *assumption*; we do not run a sensitivity analysis (Rosenbaum bounds) to bound the effect of unmeasured confounders. (iii) The "readmission" outcome in pair (b) is operationalised as `Number_Of_Placements` ≥ 2 within a single fiscal year; a longer-window definition would yield different but still identifiable effects, and a hazard-model formulation (Cox / accelerated-failure-time) would let us decompose by time-since-release rather than by within-year incidence. (iv) The χ^2 test for cross-region churn assumes independent observations; person-level clustering in repeat-placement data would inflate the χ^2 statistic relative to a cluster-robust version. (v) The 99th-percentile truncation on `NumberConsecutiveDays_Segregation` dampens the right tail; a Tobit or zero-inflated regression would give a cleaner causal contrast at the cost of distributional assumptions.

7 Conclusion

OTIS data are simultaneously consistent with all five Goffmanian predictions tested. Repeat placement is heavy-tailed ($\hat{\alpha}_{\text{churn}} = 1.618$, $G = 0.575$); time-to-readmission is dominated by within-year returns; inter-region transfer is routinised at $p < 10^{-3}$; alert-state complexity is heavy-tailed (1 098 multi-alert person-years against 58 207 no-alert person-years); and the three causal estimators converge on identifiable ATEs for all three (T, Y) pairs. The strongest empirical statement we can make is the concordance of IPW, AIPW, and DML across all nine cells of the causal grid: a mental-health alert raises subsequent suicide-risk-alert probability by ~ 0.18, high alert complexity reduces in-year readmission by ~ 0.06, and inter-region transfer raises consecutive segregation by ~ 0.33 days. These effects are small but identified, and they describe a routinised classification-and-churn apparatus that operates on a heavy-tailed minority of the correctional population — exactly the structural pattern Goffman described in qualitative form sixty-five years ago.

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